

PROFESSIONAL EXPERIENCE			
Robotspace Jun’ 25 - Present	● Built the software stack for a 4-axis Linear Robot consisting of 4 modules integrated through MQTT ● Planned the trajectory for the robot using Behaviour Trees and robot motion using Finite State Machines ● Deployed the software stack on ioWare and integrated it with FluidNC through G-Codes for Motor Control ● Implemented a Automated Planning and Scheduling system for Production Planning in factories ● Built a Cutting Plan Optimization tool using Google OR-Tools for Tower Manufacturing which reduced raw material wastage by 50% ● Deployed a PLC Based Assembly Verification system for Manufacturing Lines in Schneider Electric		
EDUCATION & SCHOLASTIC ACHIEVEMENTS			
Program	Institute	CGPA/%	Graduation
B.Tech in Aerospace Engineering	Indian Institute of Technology Madras	7.42	2025
XII (CBSE)	Sudharsanam Vidyaashram	92.2%	2021
X (CBSE)	Sudharsanam Vidyaashram	95.4%	2019
● Secured 2nd place in European Hyperloop Week for our work on Full Scale Technical Aspects. ● Ideated and Authored a Research report on the novel topic “Analysis of Shock Waves in Hyperloop System”.			
SKILLS			
Tools: ROS2, PyTorch, MQTT, Gazebo, Isaac Sim, Isaac Lab, Ansys Fluent, Fusion 360, Ansys Mechanical, AutoCAD, Latex			
Languages: Python			
PROJECTS			
Visual Pick and Place using Imitation Learning*	● Implemented DAgger training for colour based pick and place task with Franka Robot in Isaac Lab. ● Mapped camera stream input to a joint angle output using a ResNet18+LSTM+MLP neural network. ● Trained the simulation for 30 iterations with 20 rollouts and 5000 steps each		
Bachelor’s Thesis Project Guide: Satadal G	● Analyzed performance of Guidance Algorithms for Multi-Agent Formation Control using MAVROS and Gazebo ● Implemented and verified the results of the Detective Deviated Pursuit algorithm in MATLAB. ● Used Crazyflie 2.1 drones with Lighthouse Positioning System to implement the formation algorithms.		
Mechanical Head Avishkar Hyperloop Aug ’23 – July ’24	● Led a Team of 8 members to engineer and build the Mechanical Systems of the Hyperloop pod. ● Represented IIT Madras and India, securing 2nd place in European Hyperloop Week in Zurich, Switzerland. ● Proved the presence of turbulence due to SBLI in acceleration phase which travels backward and becomes steady. ● Conducted structural simulations to ensure max deformation of Chassis and Aerodynamic Shell below 0.5mm.		
Project Member Avishkar Hyperloop Mar ’23 – Aug ’23	● Performed dynamic CFD simulation on the different pod-tube configurations to analyze drag variation. ● Observed a 56% reduction in drag at 0.4 Mach in Connected Tube Configuration over single tube configuration. ● Filed a patent on the novel idea of using Connected Tube Configuration for Hyperloop systems.		
Project Head Aero Club May ’23 – Mar ’24	● Headed a Team of 5 members to design a QuadPlane for Search & Aid Operations in the ocean. ● Developed and executed Python scripts integrating MAVROS with ArduPilot SITL and Gazebo. ● Used OpenCV to detect the presence of subjects needing help and determined the required flight direction.		